AI-Assisted Music Intervention for Older Adults via Multimodal Behavior Analysis

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Introduction

Global aging is accelerating rapidly — over 1.1 billion people are aged 60+ worldwide, with depression and anxiety affecting approximately 14% of this population. Music therapy has shown significant benefits for mood, cognition, and quality of life in older adults, including those with Alzheimer's disease and related dementias (ADRD). However, clinical assessment still relies heavily on manual therapist observation, which is time-consuming and difficult to scale.

This work presents a multimodal dataset and AI pipeline designed to automatically recognize music-induced emotions and micro-behaviors in older adults, supporting scalable, personalized music-based interventions.

Dataset Overview

Item	Count / Description
Participants	17 older adults (aged 60+)
Gender distribution	12 female, 5 male
Music clips per participant	12 (30 s each)
Total clips collected	204
Clips retained after cleaning	200 (4 excluded)
Frames per clip	∼900 (30 fps × 30 s)
Total frames	≈183,220 (across 200 clips)
Feature files	2 CSVs per clip (`FacialExpression.csv`, `UpperBodySkeleton.csv`)
Annotation files	1 Excel per clip (`behavior_labels.xlsx`)
Self-report files	1 Excel per clip (`self_report.csv`)

Table 1: Dataset Summary and File Organization

Annotation Pipeline

Three-stage annotation process:

1.Therapist Observation — Board-certified music therapist reviewed all recordings, noting facial expressions, head/body movements, gaze, and mood changes in real time

2.LLM-Structured Extraction — GPT-4o processed transcripts to extract timestamped behavior-emotion mappings

3.Manual Review — Research team validated timestamps and refined annotations

4.Action Filtering — 12 canonical classes across 4 modalities

Final output:

1. Free-text annotations
2. 12-class filtered taxonomy

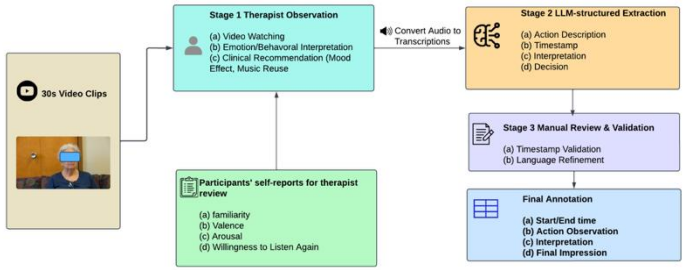


Fig 1: Annotation pipeline

Modality	Level 1	Level 2	Level 3
Facial Expression	Neutral Facial Expression	Positive Expression	Negative Expression
Eye Gaze	Look Down	Look Front	Look Away
Head Movement	No Head Movement	Repeated Head Movement	Static Head Movement
Body Movement	No Body Movement	Posture Adjustment	Body to Rhythm

Table 2: Canonical Action Classes Organized by Modality

Data Analysis

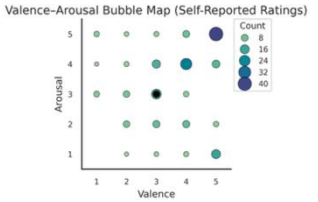


Fig 2: Distribution of self reports

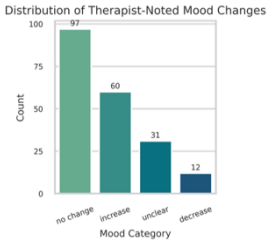


Fig 3: Distribution of mood change

self-reported valence/arousal:

- Most responses cluster in high valence
- Arousal is mainly moderate to high
- Negative response are rare
- Music stimuli effectively evoke positive and engaging emotions

Music therapist's Annotation:

- Most sessions are labeled as "no change"
- Positive mood changes are also frequently obseved
- Negative mood are rare
- Observed mood change are generally subtle and stable with a tendency toward positive or neutral responses.

Micro-Behavior Detections

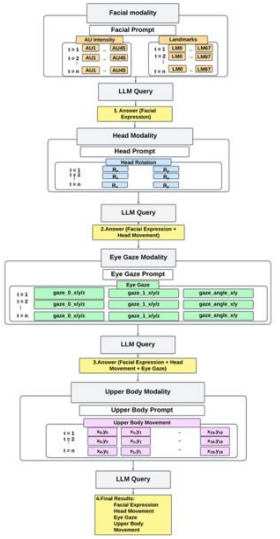


Fig 4: Overview of Chain-of-Modality baseline



Fig 5: Global action overlap between therapist annotation and LLM predictions under the 12 – class taxonomy

Modality	ROUGE-L_P	ROUGE-L_R	ROUGE-L_{F1}	Jaccard
Body	0.2273	0.9838	0.3409	0.4200
Eye	0.5497	0.7990	0.5579	0.6792
Facial	0.3948	0.8857	0.5069	0.5758
Head	**0.5903**	**0.7851**	**0.6066**	**0.6292**

Table 3 : Results of 12-class micro-action detection across four modalities

Mood Change Direction Prediction

Input Setting	Acc	Macro F1	F1 (Inc)	F1 (Not Inc)
Therapist-Annotated behaviors	0.88	0.83	0.75	0.92
CoM Predicted behaviors	0.53	0.44	0.21	0.67
ML on Raw Multimodal Signals	0.68	0.67	0.60	0.74

Table 4: Performance of Mood Dynamics Prediction

Future Directions

- Expand participant cohort and include multiple therapist annotators
- Develop real-time AI-assisted music recommendation system
- Extend to dementia-specific populations

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